SpectraQuant: From Budget-Aware Allocation Failure to Verified Clipping Transfer

A Reproducible Study of Local Compression Objectives and Whole-Model Quality

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Independent Research Project
https://github.com/OussemaHarrabi/SpectraQuant

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Abstract

SpectraQuant investigates whether inexpensive local compression objectives can preserve autoregressive language-model quality under a measured equal-byte checkpoint budget. The project began with a sensitivity-aware joint low-rank and low-bit allocator. On SmolLM2-135M, its trained arm failed to beat uniform INT4: perplexity was 23.1698 without the preparation regularizer and 23.3131 with it, versus 18.6193 for uniform INT4 at matched target bytes. A deployment-matched follow-up then evaluated the serialized and reloaded residual representation; validation again retained uniform INT4, while six alternatives worsened held-out perplexity by 5.52–14.75%. Diagnostic experiments showed that summed marginal layer gains overpredicted the quality of the jointly modified model.

This failure motivated a simpler conventional clipping control. A first frozen C4 holdout improved from 28.0056 to 27.4656 perplexity (1.9279%) at identical payload and full-checkpoint bytes. A subsequently frozen transfer study covered SmolLM2-360M and Pythia-410M on reserved C4 and PG19 prefixes. The primary weight-MSE control improved 2/4 cells and achieved a 0.5752% geometric-mean reduction, but failed its predeclared 3/4 consistency gate. Predeclared activation-weighted axis1 controls reduced C4 perplexity by 7.39–7.70% on SmolLM2 and 5.75–6.09% on Pythia at identical bytes, while regressing on Pythia PG19. The study therefore establishes a bounded C4 result and a model/domain transfer failure, not a universal compression improvement. All 52 expected Phase-4 artifacts and 42 checkpoint files were authenticated.

Keywords: transformer compression; post-training quantization; INT4; clipping; sensitivity; low-rank approximation; reproducibility; negative results

1 Introduction

Transformer compression is often optimized through local quantities: weight reconstruction error, activation-weighted error, Hessian surrogates, spectral statistics, or layer-wise calibration loss. The operational outcome, however, is global. Every modified layer participates in one autoregressive computation, and small locally favorable changes may interact poorly when composed.

SpectraQuant asks one practical question:

> *Can cheap local compression scores choose a low-bit representation that preserves end-to-end language quality at the same measured checkpoint budget?*

The project is unusual in treating a failed original method as the beginning of the investigation rather than an outcome to omit. The experimental progression is:

1. test a joint low-rank/low-bit allocator against uniform INT4;
2. correct the evaluation to score the actual packed/reloaded operator;
3. test whether marginal scores predict joint-model loss;
4. reduce the intervention to conventional clipping;
5. freeze a two-model/two-domain transfer study before reading holdouts.

The final contribution is therefore methodological and empirical. The repository implements deterministic experiment plans, equal-byte gates, packed-storage checks, cloud notebook generation, authenticated collection, generated figures, and explicit measurement classes. Empirically, it documents where local objectives fail, where clipping helps, and where that help stops transferring.

2 Background and Related Work

Post-training quantization (PTQ) maps dense weights to a small discrete codebook after training. Round-to-nearest (RTN) is a strong transparent baseline. Second-order and activation-aware methods such as GPTQ [1],

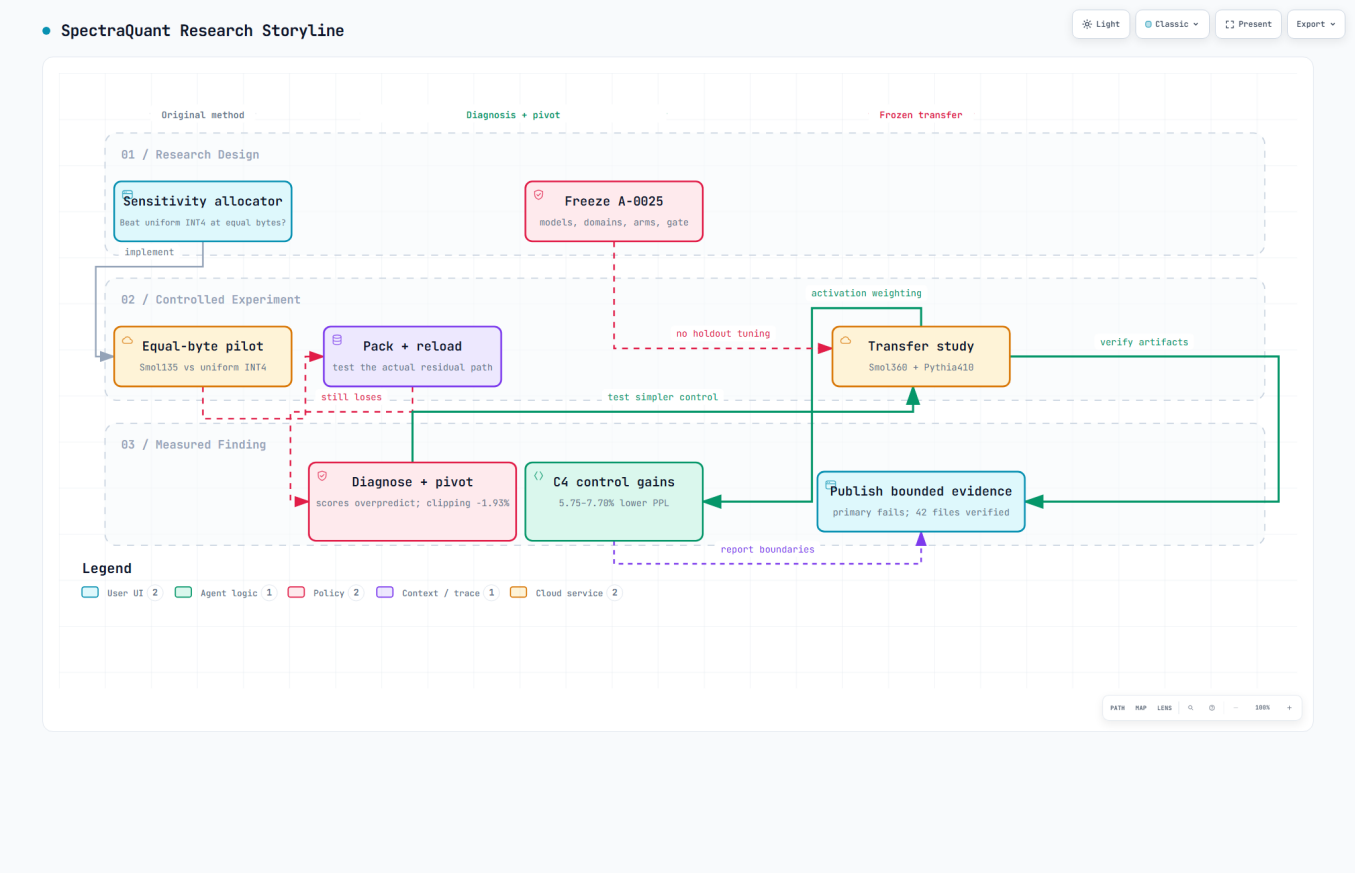


Figure 1: The coherent study progression. Later phases simplify or test an assumption exposed by the preceding failure; they are not an unrelated search for a favorable number.

AWQ [2], and SmoothQuant [3] demonstrate that calibration statistics can materially change low-bit behavior. HAWQ-style methods use sensitivity estimates to allocate precision [4].

Low-rank adaptation and quantization are also frequently combined. LoftQ initializes low-rank adapters against quantization error [5], while LR-QAT embeds low-rank structure inside quantization-aware training [6]. These works motivate the original SpectraQuant hypothesis but do not remove the need for equal-storage comparisons or deployment-matched evaluation.

The present study does not claim a new PTQ algorithm. Its strongest evidence is about *evaluation validity*: local proxy improvements and lower weight error can fail to predict whole-model perplexity, and a control that helps one architecture/domain pair can regress on another.

3 Problem Formulation

For compressible layer ℓ with weight $W_\ell \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, the original allocator chooses rank r_ℓ and bit width b_ℓ:

$$\min_{\{r_\ell, b_\ell\}} \sum_{\ell=1}^{L} \widehat{E}_\ell(r_\ell, b_\ell) \quad \text{s.t.} \quad \sum_{\ell=1}^{L} \text{bytes}_\ell(r_\ell, b_\ell) \le B. \tag{1}$$

Here $\widehat{E}_\ell$ is a local sensitivity surrogate and B is a frozen payload budget. The deployed residual form is

$$\widehat{W} = Q(W) + Q(B)Q(A), \tag{2}$$

with rank zero representing plain quantization. SpectraQuant measures the serialized target payload and full checkpoint rather than inferring equality from nominal bit widths.

The later clipping study fixes INT4/group32 quantization and chooses a clipping scale α from a predeclared grid. Weight-only MSE minimizes local reconstruction error. The activation-weighted diagonal proxy is proportional to

$$\widehat{E}_{\text{diag}} = \sum_j \mathbb{E}[x_j^2] \sum_i \left(W_{ij} - Q_\alpha(W)_{ij}\right)^2. \tag{3}$$

It omits cross-feature covariance and is not measured output error. To avoid a degenerate comparison, activation weighting is evaluated on axis1 against axis1 RTN and MSE controls; for axis0, the input second moment is constant inside each optimized output-row group.

4 Research Design

4.1 Decision discipline

Three rules govern all headline comparisons:

1. quality is compared only when measured storage matches;
2. evaluation reloads the files that the experiment actually writes;
3. model revisions, corpora, arms, gates and token budgets are frozen before fresh holdout access.

Protocol changes are append-only amendments. Failed gates remain visible. Exploratory calibration subsets are never relabeled as independent training seeds, and exposed holdouts are not reused for tuning.

4.2 Measurement classes

The repository separates four forms of evidence:

- **Class 1**: analytical size estimates;
- **Class 2**: dense execution of reconstructed quantized weights;
- **Class 3**: bytes of serialized payload/checkpoint files;
- **Class 4-CPU**: a bounded real CPU low-bit-kernel path.

Phase-4 perplexity is Class 2 and storage is Class 3. No Phase-4 number implies GPU low-bit acceleration, service throughput, or latency.

4.3 Phase-4 frozen transfer protocol

The transfer design uses SmolLM2-360M and Pythia-410M at immutable revisions. The reserved corpora are C4 validation rows 320–639 and the first 50 PG19 test books as source pools. Calibration uses the first 200 C4 training documents.

Seven symmetric INT4/group32 checkpoints are written per model: RTN and weight-MSE clipping on axis0 and axis1, plus activation-weighted axis1 clipping from calibration subsets 0, 1, and 2. The clipping grid is $\{1.00, 0.95, 0.90, 0.85, 0.80, 0.75\}$. There is no holdout NLL search and no gradient training. All checkpoints are frozen before holdout loading.

Each corpus/model cell predicts exactly 51,200 tokens with sequence length 1024 and preserved document boundaries. Comparisons are paired inside a model because the two tokenizers differ. Document-cluster bootstrap intervals describe corpus uncertainty; they are not training-seed intervals.

5 Experimental Progression

5.1 Original allocation result

The equal-memory SmolLM2-135M comparison refuted the original improvement claim. Uniform INT4 achieved 18.6193 perplexity. The trained allocated arm scored 23.1698 without the preparation regularizer and 23.3131 with it. The regularizer therefore worsened the trained arm by approximately 0.62%.

A bounded LR-QAT-direction check improved perplexity from 26.9432 to 24.1265. This closed 21.8% of the PTQ-to-FP16 gap, below the preregistered 25% threshold. It is reported as a partial directional reproduction rather than confirmation.

Table 1: Primary frozen transfer result.

Model	Corpus	RTN	MSE	Δ_{PPL}
Smol360	C4	21.8140	22.3901	-2.6408%
Smol360	PG19	16.6734	16.8819	-1.2501%
Pythia410	C4	32.3251	31.3777	+2.9311%
Pythia410	PG19	27.9244	27.0500	+3.1313%

5.2 Deployment-matched correction

Phase 2 scored the serialized and reloaded $Q(W) + Q(B)Q(A)$ operator under an equal INT4 target-payload ceiling. Validation retained uniform INT4. Six mixed-bit or residual-inclusive alternatives worsened the held-out WikiText perplexity by 5.52–14.75%. Twenty-one checkpoint files passed validation.

5.3 Joint-loss diagnosis and pivot

Phase 3 compared summed marginal predictions with composed-model loss. All six candidate allocations overstated joint calibration gains and worsened loss on 16 disjoint training blocks. This ruled out a simple explanation based only on the earlier test split.

The intervention was then simplified to conventional weight-only clipping. On a new C4 holdout, the fixed control improved perplexity from 28.0056 to 27.4656, a 1.9279% reduction, with identical 59,719,680-byte target payloads and 116,522,698-byte full checkpoints. This single-model pilot motivated the frozen transfer test; it was not treated as proof of generality.

6 Transfer Results

6.1 Primary weight-MSE result

Table 1 compares MSE clipping with RTN using the same group32/axis0 representation. Positive reduction denotes lower perplexity:

$$\Delta_{\text{PPL}} = 100 \left(1 - \frac{\text{PPL}_{\text{method}}}{\text{PPL}_{\text{RTN}}}\right). \tag{4}$$

The within-cell log-ratio geometric-mean reduction is **0.5752%**, exceeding the 0.5% magnitude requirement. Only **2/4 cells** improve, below the frozen 3/4 consistency requirement. The primary transfer hypothesis therefore fails.

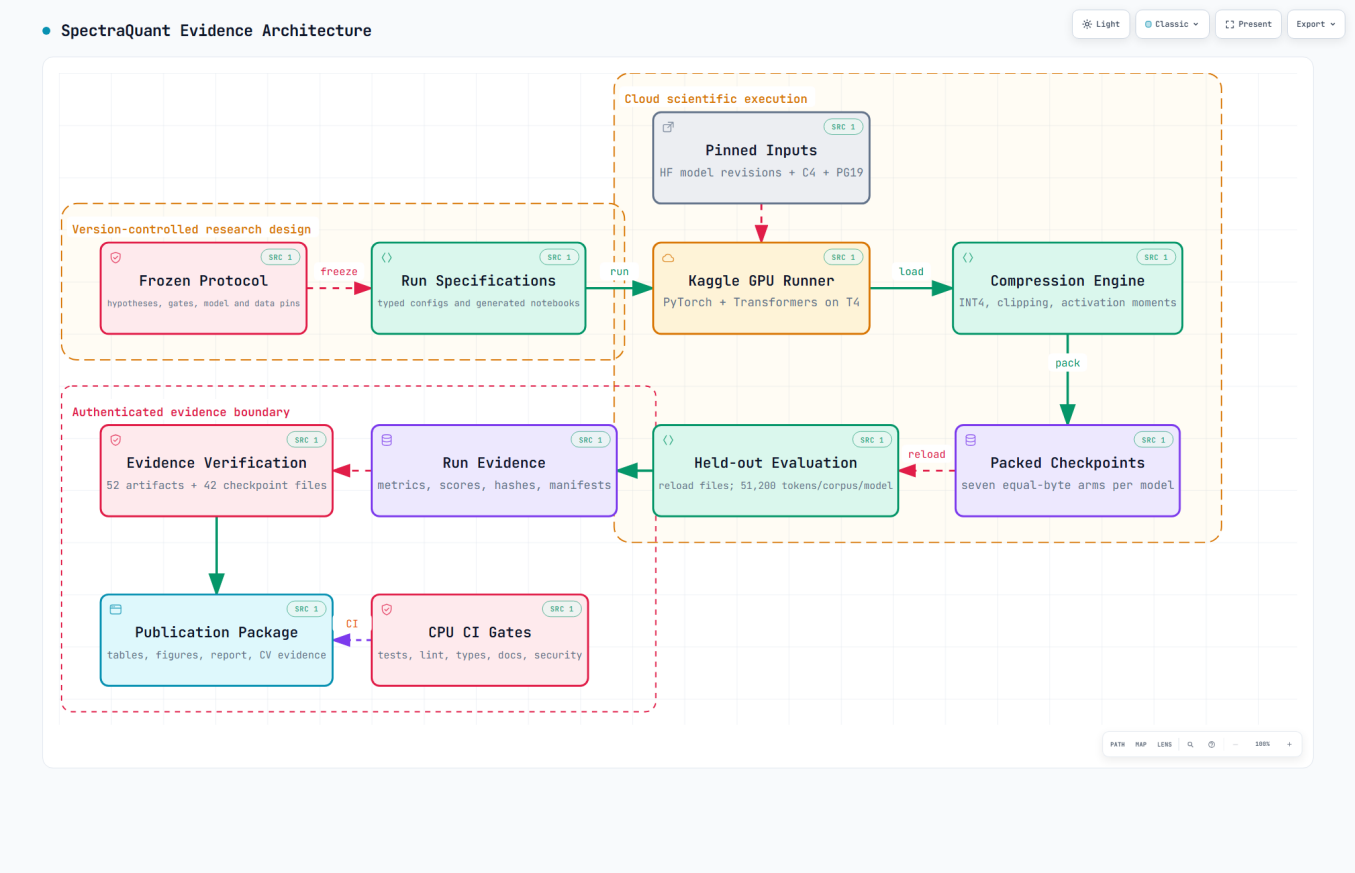


Figure 2: Evidence architecture. Frozen design and typed run specifications feed cloud execution; packed checkpoints are reloaded for evaluation; manifests, hashes and CI gates separate measured evidence from publication claims.

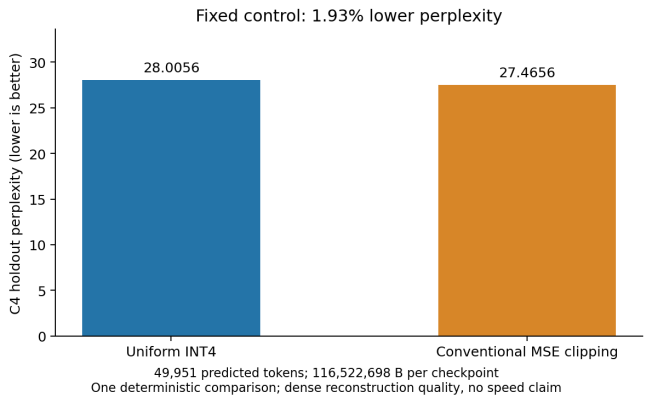


Figure 3: The first fresh-holdout clipping comparison. This bounded positive result triggered a transfer study rather than immediate generalization.

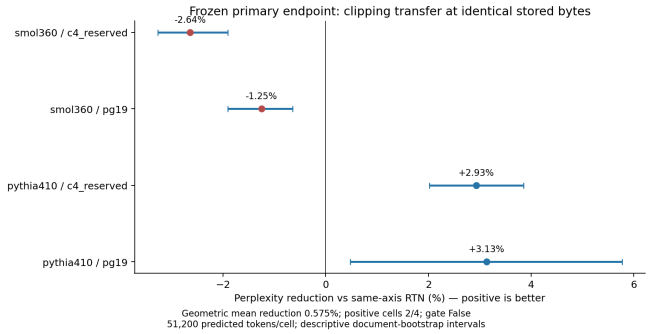


Figure 4: Primary transfer cells and descriptive document-cluster intervals.

6.2 Activation-weighted secondary control

Table 2 reports the range across all three calibration subsets. The baseline is each model's matched axis1 RTN checkpoint.

The control improves reserved C4 on both architectures and SmolLM2 PG19, but regresses on Pythia PG19. Calibration-domain sensitivity is plausible, yet this study does not isolate it causally because it does not add book-domain calibration. The supported statement is therefore a C4 improvement at equal bytes, accompanied by a documented transfer failure.

6.3 Storage equality and artifact verification

All seven arms for a model have identical storage:

The collection process verified 52 expected artifacts and 42 dynamically selected checkpoint files, including checksums, source revision, model/data fingerprints, to-

Table 2: Activation-weighted PPL reduction ranges.

Model	Corpus	Reduction range
Smol360	C4	+7.3882 to +7.7028%
Smol360	PG19	+5.8906 to +6.1849%
Pythia410	C4	+5.7462 to +6.0939%
Pythia410	PG19	-2.5608 to -1.0864%

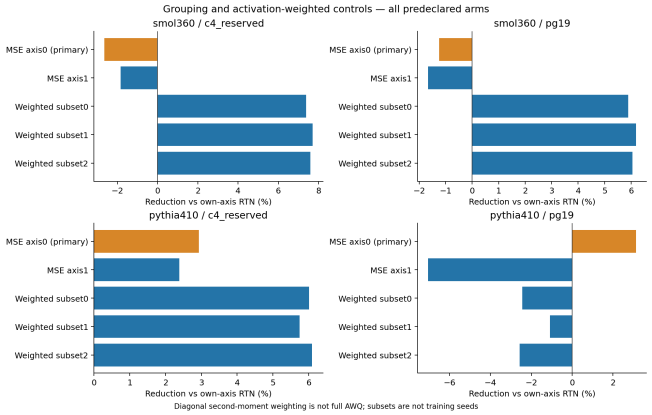


Figure 5: All predeclared clipping controls. Secondary results do not override the failed primary gate.

ken counts, and the ordering constraint that checkpoint selection precedes fresh holdout access.

7 Why Local Objectives Failed

Weight-MSE clipping reduced aggregate weight squared error by 9.98% on SmolLM2 axis0 and 10.19% on axis1, yet worsened perplexity on both corpora. Pythia axis1 supplies another counterexample: 10.69% lower weight error accompanied 7.06% worse PG19 perplexity.

The Phase-3 allocation diagnostics show the same issue from another angle. Layer effects measured in isolation were not additive after simultaneous deployment. Transformer residual paths, normalization, activation shifts and error propagation couple layers. A useful proxy must therefore be validated against composed-model outcomes rather than only local score correlation.

8 Reproducibility and Compute

The Phase-4 scientific source was frozen at commit `71a1ec11cea6be90a6ab49356e81d980f4667784`. Typed run specifications record model revisions, dataset windows, arms, token budgets, clipping grids and acceptance gates. The generated Kaggle notebooks preserve those inputs.

Two private Kaggle runs completed on their first submission. Their runner intervals correspond to approximately 1.511184 visible T4 device-hours combined, including installation but excluding export/platform overhead. This value is execution telemetry, not exact account billing.

Table 3: Measured Phase-4 storage per arm.

Model	Target payload	Full checkpoint
Smol360	176,947,200 B	271,561,031 B
Pythia410	198,844,416 B	302,615,137 B

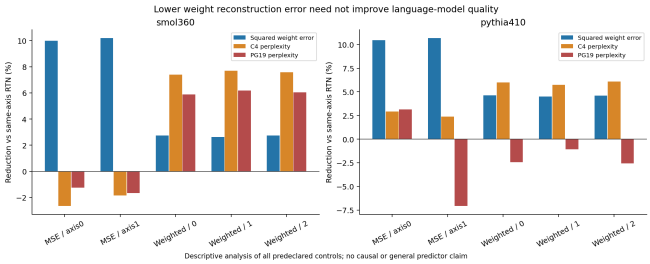


Figure 6: Lower local reconstruction error is not sufficient for lower autoregressive perplexity.

The project records Python, PyTorch, Transformers, CUDA and hardware metadata. Collection authenticates remote manifests before accepting derived tables. Scientific figures are generated from committed artifacts with input/output hashes. Local CI checks formatting, types, tests, documentation and security.

9 Limitations

- Phase-4 calibration subsets are not independent model-training seeds.
- Only two sub-billion-parameter model families and two reserved corpora are tested.
- Perplexity is measured through dense reconstruction, not a deployed low-bit GPU kernel.
- The diagonal activation proxy omits feature covariance.
- Modern PTQ baselines such as full GPTQ/AWQ implementations are reviewed but not run as matched Phase-4 arms.
- Exposed holdouts cannot support further tuning; a new hypothesis requires a new protocol and untouched data.

10 Conclusion

SpectraQuant did not validate its original joint low-rank/low-bit allocator. Correcting deployment evaluation and inspecting joint loss strengthened that negative result. The subsequent pivot showed that conventional clipping can deliver meaningful equal-byte gains, but also that transfer depends on model, corpus and grouping. Activation-weighted clipping produced repeatable C4 improvements across both tested architectures and all three calibration subsets, while failing on Pythia PG19.

The main result is therefore not a universal compres-

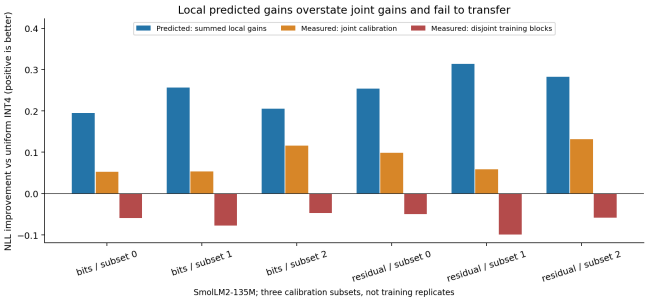


Figure 7: Marginal allocation predictions versus joint-model behavior.

sion recipe. It is a reproducible account of how to turn a failed hypothesis into useful engineering knowledge: measure the deployed representation, freeze the next test, retain negative cells, authenticate artifacts, and bound every claim by the evidence that actually ran.

Artifact Availability

Code, protocols, generated notebooks, authenticated bounded evidence, figures, and report source are available at `https://github.com/OussemaHarrabi/SpectraQuant`.

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